NAME: ASHOK

REG.NO: 192312558

Ex 4: Crypt-Arithmetic Problem (SEND + MORE = MONEY)

CODE:

```
import itertools


def solve_cryptarithmetic():
  letters = 'SENDMORY'
  for perm in itertools.permutations(range(10), len(letters)):
    s, e, n, d, m, o, r, y = perm
    if s == 0 or m == 0:
      continue
    send = s*1000 + e*100 + n*10 + d
    more = m*1000 + o*100 + r*10 + e
    money = m*10000 + o*1000 + n*100 + e*10 + y
    if send + more == money:
      return {'S': s, 'E': e, 'N': n, 'D': d, 'M': m, 'O': o, 'R': r, 'Y': y}


res = solve_cryptarithmetic()
print("Solution found:")
for k, v in res.items():
  print(f"{k} = {v}")
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-4.py ======
Solution found:
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
```